

Triple-drug fixed-dose combination of Sitagliptin + Glimepiride + Metformin: New addition in the Anti-diabetic armamentarium

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Background

Prevalence of T2DM in India^{1,2}



T2DM in Asian Indian people is characterized by:³



Young age of onset



Low levels of BMI

Rationale for combination

- Less than 50% patients with T2DM achieve their recommended glycemic goals⁴
- Among Indian patients with diabetes, mean HbA1c is 8.9%⁵
- T2DM is progressive disease and patients require multiple antidiabetic agents with differing MOAs to achieve target HbA1c⁶
- High pill burden and complex treatment regimens decrease adherence to therapies⁴
- Each 10% increase in compliance is associated with a 0.1% HbA1c reduction⁷

What are guideline recommendations for management of T2DM?

2020: RSSDI-ESI recommendations⁸

- Consider initiating combination therapy if HbA1c > 1.5 above the target initial therapy

2023: AACE recommendations⁹

- For those with an initial HbA1c of >9% or 1.5% above goal, then 2 or 3 antihyperglycemic agents should be initiated concomitantly

2023: ADA guidelines¹⁰

- Pharmacologic approaches that provide adequate efficacy to achieve and maintain treatment goals should be considered, such as Metformin or other agents, including combination therapy

Sitagliptin, Metformin and Glimepiride in T2DM

Sitagliptin



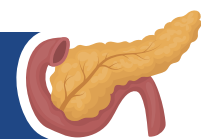
- Inhibits DPP-4 enzyme and thereby prevents degradation of GLP-1¹¹
- This causes an increase in insulin release after meals and improved glucose tolerance¹¹
- **Demonstrated better response in Asians¹²**

Metformin



- Suppresses hepatic glucose output, reduction of insulin resistance and increase in GLP-1 release¹³
- **First-line therapy as per all guidelines^{8,9,14}**

Glimepiride



- Stimulate insulin release from pancreatic β -cells¹⁵
- Lower risk of hypoglycemia compared to other SUs¹⁶
- **Very potent and commonly used in India to achieve glycemic targets¹⁷**

Addition of Sitagliptin to Metformin while optimizing Glimepiride dose to improve glycemic control in T2DM

Multinational, randomized, double-blind, parallel-group study (n=194)¹⁸

Addition of Sitagliptin to Glimepiride + Metformin reduced HbA1c by

0.89%

Prospective, 52-week, single centre, single arm, intervention study (n=39)¹⁹

Addition of Sitagliptin to Glimepiride with/without Metformin reduced HbA1c by

0.81%, $p < 0.001$

Further, addition of Sitagliptin to combination therapy:¹⁹

- Reduces dose of Glimepiride and preserves β -cell function
- Reduces incidence of hypoglycemia

Safety of Sitagliptin, Metformin and Glimepiride in T2DM

Sitagliptin + Metformin + Glimepiride^{16,20}



Metformin

→ Proven CV safety²¹

Sitagliptin

→ Proven CV safety through large TECOS study²²

Glimepiride

→ Proven CV safety in CAROLINA study²³

- **Well-tolerated**
- **Weight neutral**

Sitagliptin + Metformin + Glimepiride: Impact across value chain of T2DM management

Patient impact^{17,18,19}

- Better HbA1c control
- Improves β -cell function
- Low risk of hypoglycemia
- Improves compliance

Healthcare system impact²⁴

- Decrease in healthcare resource utilization and strain on healthcare
- Reduces/delays long-term complications

Community impact²⁴

- Decrease in economic impact
- Improves quality of life

For chronic conditions such as T2DM, this option has the potential to optimize compliance, and may enhance patient convenience and ease of use which may translate into better treatment outcomes

Conclusion

Sitagliptin + Glimepiride + Metformin in fixed-dose combination has the potential to provide better glycemic control without increased risk of hypoglycemia, improve treatment compliance and help patients to achieve their glycemic targets, thereby reducing long-term complications.

Abbreviations:

AACE: American Association of Clinical Endocrinologists; ADA: American Diabetes Association; BMI: Body mass index; CV: Cardiovascular; DPP-4: Dipeptidyl peptidase-4; GLP-1: Glucagon-like peptide 1; HbA1c: Glycated hemoglobin, MOA: Mode of action; RSSDI-ESI: Research Society for the Study of Diabetes in India-Endocrine Society of India; SU: Sulfonylureas; T2DM: Type 2 diabetes mellitus; TECOS: Trial Evaluating Cardiovascular Outcomes with Sitagliptin

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